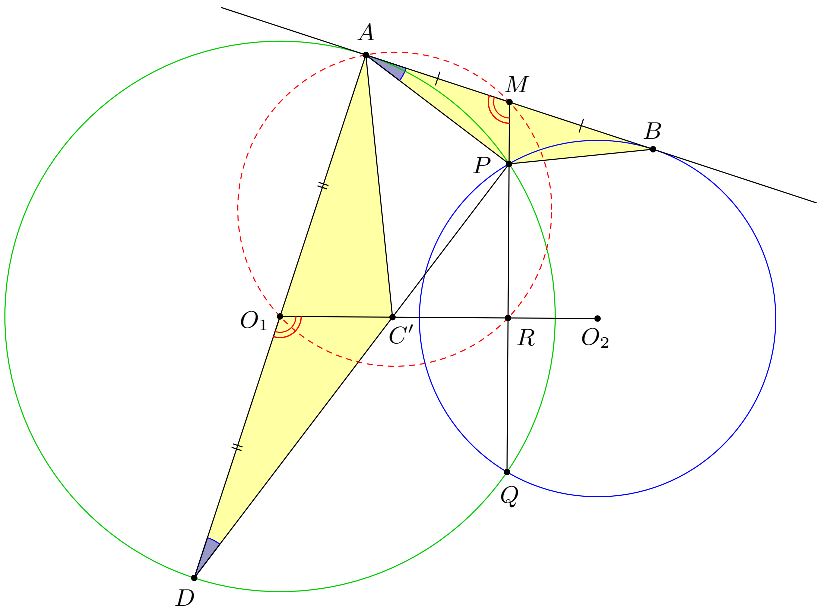
IMO 2020
61st International Mathematical Olympiad Saint Petersburg Russia

HK IMO Training 2019-2020
Problem Set 31 Suggested Solutions

Problem 121. Let P be one of the intersection points of two circles with centres O_1, O_2. A common tangent touches the circles at A, B respectively. Let the perpendicular from A to the line BP meet O_1O_2 at C.

Prove that AP is perpendicular to PC.

Solution.



Let D be the antipodal point of A on the circle with centre O_1. Join PD and let it meet O_1O_2 at C'. Then $AP \perp PC'$. If we can prove that $BP \perp AC'$, this will show that $C' = C$.

Let the two circles meet again Q. Extend PQ so that it meets AB at M. Then it is well-known that $AM = MB$ (or simply, $AM^2 = MP \times MQ = MB^2$). Let PQ meet O_1O_2 at R. Then clearly, A, O_1, R, M are concyclic, and so $\angle AMP = \angle DO_1C'$. Since $\angle MAP = \angle O_1DC'$, we have

$$\triangle AMP \sim \triangle DO_1C'.$$

Now, since $DA = 2DO_1$ and $AB = 2AM$, we have

$$\triangle ABP \sim \triangle DAC',$$

and so

$$\angle ABP = \angle DAC' = 90^\circ - \angle C'AB.$$

Therefore $BP \perp AC'$ and the result follows. □

Problem 122. In an acute-angled triangle ABC with $\angle CAB > 60^\circ$, let

- H be its orthocentre;
- M, N be two points on AB, AC respectively, such that $\angle HMB = \angle HNC = 60^\circ$;
- O be the circumcentre of $\triangle HMN$;
- D be a point on the same side with A of BC such that $\triangle DBC$ is an equilateral triangle.

Prove that H, O, D are collinear.

Solution. Let T be the orthocentre of $\triangle HMN$, $P = HM \cap AC$, and $Q = HN \cap AB$. Then

$$\angle PMQ = \angle HMB = 60^\circ = \angle HNC = \angle PNQ$$

and so P, M, N, Q are concyclic.

Since $\angle THM = \angle OHN$, we have

$$\angle PQH - \angle OHN = \angle NMH - \angle THM = 90^\circ$$

and so $HO \perp PQ$.

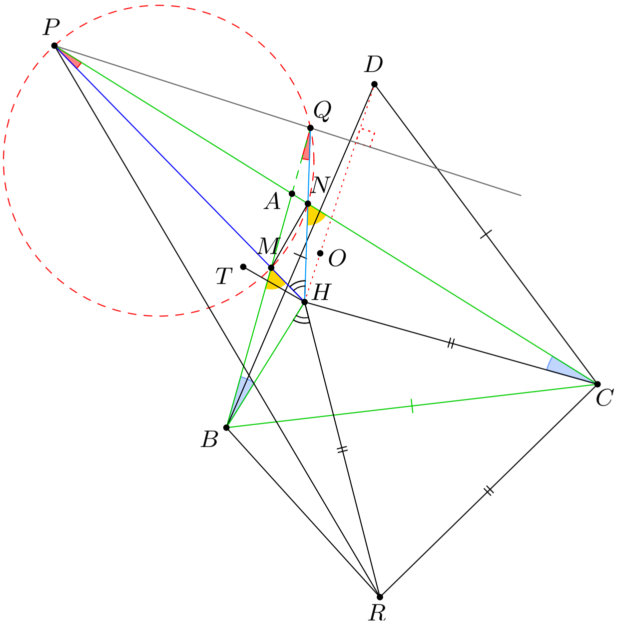
Let R be reflection of C about HP. Then $HC = HR$. Since

$$\angle HPC + \angle HCP = (\angle BAC - 60^\circ) + (90^\circ - \angle BAC) = 30^\circ$$

we have $\angle CHR = 60^\circ$ and so $\triangle HRC$ is equilateral.

Since $\angle HPC = \angle HQB$ and $\angle HCP = \angle HBQ$, we have

- $\triangle PHC \sim \triangle QHB$,
- $\triangle PHR \sim \triangle QHB$,
- $\triangle QHP \sim \triangle BHR$.



Since $\angle PHR = 150^\circ$, we have $\angle(PQ, RB) = \angle(HP, HR) = 150^\circ$. Now, since $\triangle BCD$ and $\triangle HRC$ are equilateral, we have $\triangle BRC \cong \triangle DHC$, and so $\angle(RB, HD) = \angle(CR, CH) = -60^\circ$. Hence

$$\angle(PQ, HD) = \angle(PQ, RB) + \angle(RB, HD) = 150^\circ - 60^\circ = 90^\circ.$$

Therefore $DH \perp PQ$ and so H, O, D are collinear. $\square$

Problem 123. Find all polynomials $P(x)$ with integer coefficients such that, for all integers a and b, $P(a+b) - P(b)$ is a multiple of $P(a)$.

Solution. The answers are $P(x) = cx$ or $P(x) = c$ for some integer c.

Fix an integer b and write $H(x) = P(x+b) - P(b)$. If H is identically 0, then $P(x)$ is a constant function. Substituting $P(x) = c$, the condition says that c divides 0, which is true. So constant functions work. For the remainder, we will assume that $P(x)$ is non-constant. In that case, $H(x)$ and $P(x)$ have the same leading term, so we can write

$$H(x) = P(x) + r(x),$$

where $r(x)$ is either identically 0 or $\deg r(x) < \deg P(x)$. For any integer a, we have that $H(a) = P(a) + r(a)$ is a multiple of $P(a)$. In particular, for any integer a, $r(a)$ is a multiple of $P(a)$. The degree of $r(x)$ is less than the degree of $P(x)$ (or $r(x)$ is identically 0), so we can choose some M such that for all integers $a \geqslant M$, $|P(a)| > |r(a)|$. Since $r(a)$ is a multiple of $P(a)$ for each such a, this implies that $r(a) = 0$ for all integers $a \geqslant M$. Therefore, $r(x)$ is a polynomial with infinitely many zeros, and must be identically 0. Hence $H(x) = P(x)$ and

$$P(x+b) - P(b) = P(x).$$

Since b was an arbitrary integer, this equation holds for all integers b and real numbers x. Plugging in $x = b = 0$ yields $P(0) = 0$. Plugging in $x = 1, b = 1$ gives us $P(2) = 2P(1)$. Then plugging in $x = 2, b = 1$ gives us $P(3) = 2P(1) + P(1) = 3P(1)$. Continuing this way by induction, plugging in $x = k, b = 1$ gives us

$$P(k+1) = kP(1) + P(1) = (k+1)P(1)$$

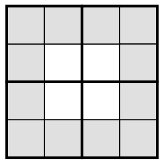
Therefore, $P(x) - xP(1)$ is a polynomial with infinitely many zeros, and must be identically 0. So if $P(x)$ is not constant, $P(x) = xP(1)$ is a linear function with no constant term. Therefore, the only non-constant solution is $P(x) = cx$ for some constant c. Substituting $P(x) = cx$, the given condition says that $ca + cb - cb = ca$ is a multiple of ca, which is true. So both of these classes of functions work and our solution is $P(x) = cx$ or $P(x) = c$ for some integer c. $\square$

Problem 124. Let $n > 1$ be an even positive integer. A $2n \times 2n$ grid of unit squares is given, and it is partitioned into n^2 contiguous 2×2 blocks of unit squares. A subset S of the unit squares satisfies the following properties:

(1) for any pair of squares A, B in S, there is a sequence of squares in S that starts with A, ends with B, and has any two consecutive elements sharing a side; and

(2) in each of the 2×2 blocks of squares, at least one of the four squares is in S.

An example for $n = 2$ is shown below, with the squares of S shaded and the four 2×2 blocks of squares outlined in bold.



In terms of n, what is the minimum possible number of elements in S?

Solution. The answer is $\frac{3}{2}n^2 - 2$, where n is an even positive integer.

Let $n = 2k$, where k is a positive integer. We will show that the minimum number of elements in S is $6k^2 - 2$. We begin with a lemma.

Lemma. Let k be a positive integer. If we have a $(2k+1) \times (2k+1)$ grid of squares, with some squares shaded so that all shaded squares are connected by side, as in condition (1) in the problem, and no 2×2 subgrid is entirely unshaded, then the number of shaded squares is at least $2k^2 - 1$.

Indeed, place the $(2k+1) \times (2k+1)$ grid on the coordinate plane with each square having area 1. The shaded squares together form a polygon whose area is equal to the total number of shaded squares. Since no 2×2 subgrid is unshaded, all of the interior vertices of the grid are inside or on the boundary of the polygon; there are $4k^2$ such vertices. By *Pick's Theorem*, we have that the area is

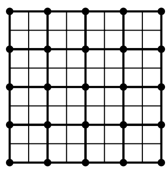
$$\frac{B}{2} + I - 1,$$

where B is the number of boundary vertices and I is the number of interior vertices. As $B + I \geqslant 4k^2$, we have

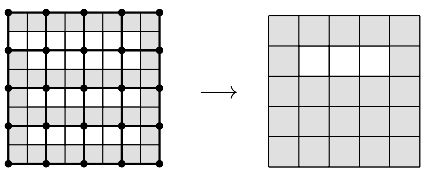
$$\frac{B}{2} + I - 1 \geqslant \frac{B+I}{2} - 1 \geqslant \frac{4k^2}{2} - 1 = 2k^2 - 1,$$

as desired.

Now returning to the problem, let S be a valid subset of the original grid, and we will show that it contains at least $6k^2 - 2$ squares. Consider the set of vertices of the n^2 blocks of 2×2 squares.

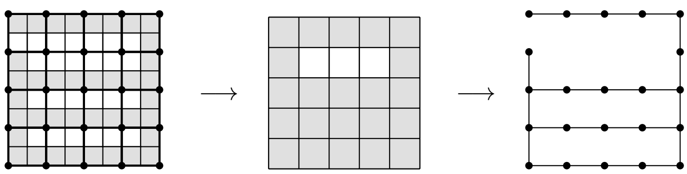


There are $(2k+1)^2$ such vertices. Construct a $(2k+1) \times (2k+1)$ grid of squares, and call it G, so that each square corresponds to one of these $(2k+1)^2$ vertices. Shade a square in G if the corresponding vertex touches a square of S. An example of such a grid and shading is shown below.



By (1), the shaded squares in G are connected by side, and by condition (2) no 2×2 block of squares in G can be fully unshaded. Therefore, by the Lemma we have at least $2k^2 - 1$ shaded squares in G.

Next, construct a graph G' as follows: the vertices correspond to the shaded squares in G. Whenever we have two adjacent squares in the same 2×2 block both in S, we draw an edge in G' connecting the two vertices that the two squares in S touch. An example graph is shown below.

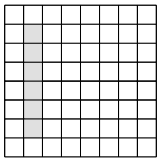


By (1), G' is a connected graph. Let $S(G')$ be a spanning tree of G'. Since G' has at least $2k^2-1$ vertices, we know that $S(G')$ has at least $2k^2-2$ edges.

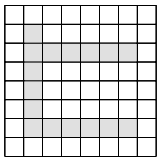
Notice that each edge in $S(G')$ corresponds to two unit squares in the same 2×2 block both being shaded. Since $S(G')$ contains no cycles, each edge corresponds to a unique such pair of unit squares. We use this to place a lower bound on the total number of shaded squares in the original grid.

In the original grid, we require at least one shaded square in each of the $4k^2$ blocks. Also, an additional shaded square in the original grid is required for each edge of $S(G')$. There are at least $2k^2-2$ edges in $S(G')$, so in total we have at least $6k^2-2$ shaded squares.

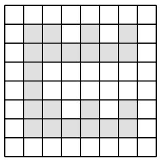
We now exhibit a construction of $6k^2-2$ shaded squares. In the accompanying diagrams, we demonstrate the construction for $n=4$. Number the rows and columns from 1 to $4k$, and denote the square in row i column j as (i,j). Shade in the $4k-2$ squares $(i,2)$ for $2\leqslant i\leqslant 4k-1$.



Next, for the k values of i with $1\leqslant i\leqslant 4k$ and $i\equiv 2 \pmod 4$, shade in $4k-3$ squares (i,j) for $3\leqslant j\leqslant 4k-1$.



Finally, shade in $k(2k-1)$ squares of the form $(2i-1,4j-1)$, for $2\leqslant i\leqslant 2k$ and $1\leqslant j\leqslant k$.



The total number of shaded squares is

$$4k-2+k(4k-3)+k(2k-1)=4k^2-3k+4k-2+2k^2-k=6k^2-2=\frac{3}{2}n^2-2.$$

as desired. $\square$